

Machine Learning in Applications

GU_02 On Edge Motion Recognition

Group 3 Project Report

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Abstract—The deployment of Machine Learning algorithms on edge devices for Artificial Intelligence of Things (AIoT) applications requires a strict balance between high accuracy and ultra-low power consumption. Neuromorphic computing, particularly Spiking Neural Networks (SNNs), offers a promising solution by mimicking the event-driven nature of biological brains. This project investigates the implementation of a Human Activity Recognition (HAR) system on the Brainchip Akida 1000 neuromorphic processor. Using accelerometer and gyroscope data from the WISDM dataset, we designed a Convolutional Neural Network (CNN) tailored to strictly adhere to the hardware’s architectural constraints. The model underwent Quantization-Aware Training (QAT) and was subsequently converted into an SNN. We evaluated the system on two scenarios: a 7-class subset (to benchmark against existing literature) and a broader 18-class dataset. Tested directly on the physical Akida hardware, the network achieved an accuracy of approximately 91.5% on the 7-class subset, closely matching the 92.5% baseline established in prior neuromorphic studies, with zero accuracy drop between the software simulator and the physical counterpart. The results demonstrate that, despite the severe constraints required for hardware compatibility, SNNs on edge accelerators can successfully perform complex HAR tasks with highly competitive accuracy.

I. INTRODUCTION

The rapid expansion of the Artificial Intelligence of Things (AIoT) has driven the need for intelligent devices capable of processing data locally. This paradigm shift towards “edge computing” mitigates issues related to latency, data privacy, and bandwidth overload, which are typical of cloud-based AI solutions. However, deploying complex Machine Learning (ML) models on edge devices introduces severe challenges regarding power consumption, memory footprint, and computational resources.

In this context, Human Activity Recognition (HAR) using wearable devices with sensors (such as Watches with accelerometers and gyroscopes) has emerged as a critical application for healthcare, fitness tracking, and ambient assisted living. Continuous monitoring of human activities generates continuous streams of data, making traditional power-hungry Artificial Neural Networks (ANNs) ill-suited for battery-powered wearables. Neuromorphic computing addresses this bottleneck by employing Spiking Neural Networks (SNNs), which process information through asynchronous, binary events (spikes). This approach dramatically reduces dynamic power consumption, offering a highly efficient alternative for on-edge HAR applications.

Despite the theoretical advantages, deploying SNNs on physical neuromorphic hardware requires bridging the gap

between standard Deep Learning frameworks and strict hardware constraints. This report presents a complete pipeline to design, quantize, and deploy a HAR model on the Brainchip Akida 1000 [1], a commercial neuromorphic System-on-Chip (SoC). We utilized the WISDM dataset [2] to train a hardware-compliant Convolutional Neural Network (CNN) baseline, which was then converted into an SNN via Quantization-Aware Training (QAT). The model was physically tested on the Akida board, validating its performance on a 7-class subset, to anchor our results against existing literature [3], and a more complex 18-class scenario.

The remainder of this report is organized as follows: Section II provides the background on neuromorphic computing and HAR; Section III details the dataset preparation, the hardware constraints, the rigorous trial-and-error network design process, and the training Pipeline and Hyperparameter Optimization; Section IV discusses the quantization and the final performance on the physical Akida hardware; finally, Section V outlines our conclusions and potential future works.

II. BACKGROUND

This section introduces the fundamental concepts underpinning the project, outlining the challenges of executing Deep Learning models on edge devices and presenting neuromorphic computing as a viable, energy-efficient alternative.

A. Human Activity Recognition and Edge Computing

Human Activity Recognition (HAR) involves the analysis and classification of time-dependent signals related to human actions. Over the last decade, the widespread diffusion of smart devices has caused wearable sensors to attract the most attention in HAR research. These devices play a crucial role in modern Artificial Intelligence of Things (AIoT) domains, ranging from continuous healthcare monitoring to smart environments and fitness tracking.

However, typical wearable sensors are inherently affected by severe limitations in terms of power. Traditional machine learning approaches often rely on transmitting acquired data to remote cloud servers for processing, which introduces critical limitations regarding latency, privacy, and continuous connectivity. To achieve the ultimate goal of edge computing for wearable devices, where data is processed locally, a paradigm shift is required to significantly reduce computational efforts. Standard Deep Neural Networks (DNNs), while highly accurate, involve dense, continuous matrix multiplications that drain the limited battery life of smart wearable devices, making them ill-suited for continuous on-edge HAR.

B. Neuromorphic Computing and Spiking Neural Networks

To mitigate the energy constraints of edge devices, neuromorphic computing provides an alternative paradigm based on biologically inspired architectures. Specifically, Spiking Neural Networks (SNNs) [4] represent a valuable candidate for energy-efficient solutions thanks to their event-based, asynchronous operations.

In biological brains, neurons communicate by spreading information in the form of discrete events called action potentials, or spikes. SNNs mimic this mechanism: computation is driven entirely by these binary spikes rather than continuous floating-point values. This temporally sparse activity naturally provides low-power computation, as energy is consumed only when and where a spike is generated and routed. Furthermore, relying on this spike-based activity, SNNs feature an intrinsic suitability for processing temporal information, making them naturally adapted for the time-series sensor data encountered in HAR tasks.

C. Neuromorphic Hardware: Brainchip Akida

While SNNs offer theoretical energy savings, realizing these benefits in real-world scenarios requires dedicated neuromorphic hardware capable of executing asynchronous operations natively. The Brainchip Akida 1000 [1] is a commercial neuromorphic System-on-Chip (SoC) designed specifically to accelerate event-based processing at the edge.

Unlike traditional Deep Learning accelerators that rely on dense arrays and heavy external memory access, the Akida architecture utilizes localized on-chip memory (SRAM) and specialized Neural Processors that compute only upon receiving non-zero spike events. By converting a natively trained Keras CNN into an event-driven SNN framework, the Akida SoC can perform complex spatial-temporal feature extraction with a fraction of the power footprint, fulfilling the strict requirements of on-edge AIoT applications.

III. MATERIALS AND METHODS

Deploying a Convolutional Neural Network (CNN) on a physical neuromorphic architecture requires a rigorous design strategy. The architecture of the Brainchip Akida 1000 (1.0) processor imposes strict hardware constraints on the topology, layer types, and parameter sizes. In this section, we describe the dataset preparation and the specific architectural choices forced by the hardware mapping process.

A. Dataset Preparation

The research utilizes the WISDM (Wireless Sensor Data Mining) dataset [2] for Human Activity Recognition. While the dataset provides inertial recordings from both smartphones and watches, our study exclusively focuses on the watch data, as wrist-worn devices are highly representative of typical on-edge wearable applications. For our primary experiments, we selected a subset of 7 hand-oriented activities (to benchmark against existing literature) and extended the analysis to an 18-class subset to stress-test the model's capacity. The raw accelerometer and gyroscope data were sampled at 20 Hz. We

segmented the continuous data streams into temporal windows of 2 seconds (overlap 1 second), resulting in 40 samples per window. Each sample contains 6 channels (X, Y, and Z axes for both the accelerometer and the gyroscope). Consequently, the native input tensor shape for a single observation is 40×6 . Data standardization was applied before feeding the inputs to the network.

B. Neuromorphic Hardware Constraints and Network Design

To achieve high accuracy on the target HAR task, we initially aimed to leverage 2D Convolutional layers. The theoretical approach involved restricting the 2D space to extract spatio-temporal features, followed by dimensionality reduction via pooling, a Flatten layer, and final Dense layers. However, the Akida 1.0 hardware presents several limitations that forced significant deviations from standard CNN design, leading to a trial-and-error optimization process:

- **Symmetric Padding and Spatial Shift:** The hardware Convolutional Neural Processors (CNPs) on the Akida 1.0 exclusively support symmetric padding and square odd kernel dimensions (eg. 1×1 , 3×3). The native 40×6 (even dimensions) input, when combined with Keras' `padding='same'`, forced mathematically asymmetric padding (e.g., zero padding on top, one on bottom). During the `cnn2snn` mapping, the compiler forced symmetric padding on the hardware, causing a silent spatial shift of the features at every layer, effectively rendering the network "blind". To solve this, we pre-padded the dataset to odd dimensions (41×9), allowing Keras to natively compute perfect symmetric padding that exactly matches the hardware logic.
- **Dimensionality Reduction without Pooling:** Standard dimensionality reduction techniques like Max Pooling and Average Pooling are not natively supported by the Akida 1.0 hardware. While the software documentation sometimes accepts them, they resulted in compilation errors or were left as "software fallbacks", violating the requirement for a fully edge-accelerated inference. Consequently, we relied entirely on strided convolutions (with a 2×2 stride) to progressively reduce the spatial dimensions of the feature maps before a standard Flatten layer.
- **Activation and Normalization Anomalies:** The hardware strictly requires bounded activation functions; standard ReLU activations triggered explicit compilation errors and were replaced with ReLU6 (capped at 6). Furthermore, while Batch Normalization is theoretically supported, we observed that it often caused silent failures during the conversion phase fusing improperly with the convolutional weights and leading to seemingly random accuracy drops on the physical board. Therefore, Batch Normalization layers were entirely omitted.
- **On-chip SRAM Optimization via 1×1 Convolutions:** To maximize energy efficiency and achieve optimal on-edge performance, the entirety of the network's parameters and activations must reside within Akida's on-chip SRAM, strictly maintaining external memory usage at

zero. During the architectural design, we identified that the primary bottleneck consuming this limited memory was the massive weight matrix generated by connecting a large Flatten layer to the subsequent Dense layer. Maintaining a high number of filters in the spatial convolutional layers naturally produced a highly dimensional flattened array, which in turn caused the Dense connections to easily exceed the available on-chip SRAM. To solve this memory constraint without sacrificing the feature-extraction capabilities of the spatial convolutions, we drew inspiration from the GoogLeNet [5] and introduced a pointwise convolution (1×1 kernel) immediately before the Flatten operation. This layer acts as a cross-channel dimensionality reduction bottleneck, effectively compressing the depth of the feature maps while preserving spatial information. By dramatically shrinking the output size of the Flatten layer, we drastically reduced the parameter footprint of the Dense connections. This structural optimization allowed us to keep a high number of filters in the preceding convolutional layers while keeping the entire model safely within the strict boundaries of Akida's on-chip SRAM.

C. Training Pipeline and Hyperparameter Optimization

The development and deployment of the models followed a strict three-phase pipeline: standard floating-point CNN training, quantization with Quantization-Aware Training (QAT), and final hardware mapping. Each phase presented unique challenges, particularly regarding the alignment between software simulations and physical hardware execution.

Phase 1: CNN Baseline Training and Optuna Optimization

The initial phase involved training a standard Keras CNN with float32 weights. The training dynamics and accuracy for both models are shown in Figures 1 and 2. To maximize the baseline performance before the restrictive quantization process, we employed the Optuna framework for automated hyperparameter optimization.

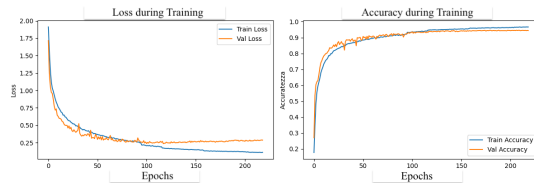


Fig. 1. Training accuracy/loss curves for the Small model.

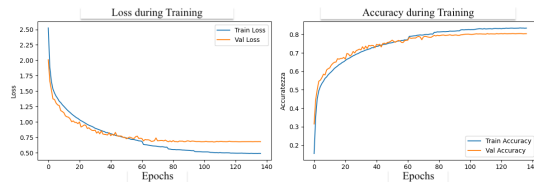


Fig. 2. Training accuracy/loss curves for the Big model.

For the 18-class (Big) scenario, Optuna successfully identified an optimal architectural and hyperparameter configuration.

The best-performing topology consisted of four convolutional layers (with 32, 128, 192, and 32 filters respectively) and a single Dense layer of 96 units, trained with a learning rate of approximately 0.00043, a dropout rate of 0.4, and a batch size of 64.

Conversely, for the 7-class (Small) scenario, while extensive Optuna trials yielded highly competitive metrics (reaching $\sim 91\%$ accuracy), none of the automated configurations managed to surpass our manually engineered architecture. The manual configuration outperformed the automated ones during the final physical execution (yielding 91.41% on the Akida board). The final CNN topologies for both scenarios are illustrated in Figures 3 and 4.

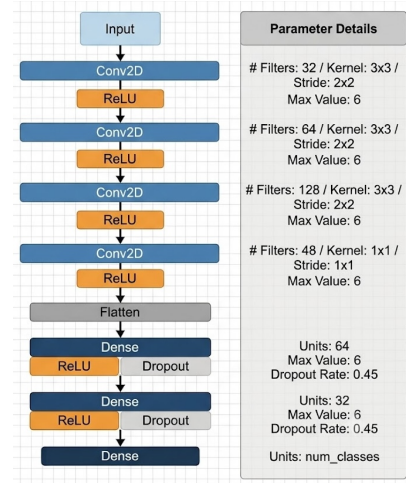


Fig. 3. Architecture of the manually optimized CNN for the 7-class (Small) scenario.

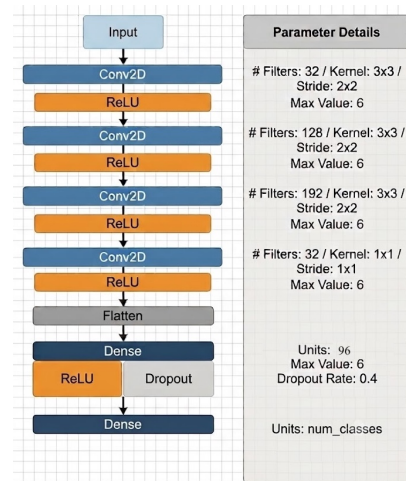


Fig. 4. Architecture of the Optuna-optimized CNN for the 18-class (Big) scenario.

Phase 2: Quantization and Quantization-Aware Training (QAT)

Once the float32 baselines were established, we utilized Brainchip's QuantizeML library to discretize the network. To preserve the integrity of the raw sensory data, the first layer was quantized to 8-bit inputs, 4-bit weights and 4-bit

TABLE I
SYSTEM PERFORMANCE AND ACCURACY EVOLUTION ACROSS DEPLOYMENT STAGES

Scenario	# Params	Memory Footprint (MB)	Keras Float32 Accuracy	Pre-QAT (Naive) Accuracy	Post-QAT Accuracy	Akida Hardware Accuracy (SNN)	Throughput (FPS)
Small (7 Classes)	138,653	0.54	94.55%	20.58%	91.18%	91.41%	2,012.03
Big (18 Classes)	304,342	1.16	80.22%	14.95%	72.81%	72.82%	2,649.21

activations (8/4/4), while all subsequent layers were strictly quantized to 4/4/4 (AKIDA Hardware limit). As anticipated, the naive quantization induced a severe "quantization shock," collapsing the accuracy of both models. To recover this lost performance, a second training phase (QAT) was necessary. During QAT, the network learns to adapt its weights within the discrete 4-bit boundaries. For the Big model, Optuna was deployed again specifically to tune the QAT hyperparameters, finding an optimal learning rate of ~ 0.00098 and a batch size of 64.

Phase 3: SNN Conversion and Hardware Mapping

The final step involved utilizing the `cnn2snn` compiler to convert the QAT-optimized model into a native Spiking Neural Network format (`.fbb`) for the physical Akida board.

Despite utilizing the official framework for the entire pipeline, this conversion phase proved to be the most critical and time-consuming bottleneck of the project. The success of the architectural choices (such as perfect symmetric padding and the omission of Batch Normalization discussed in Section 3.2) could only be fully validated after enduring both the initial training and the QAT phases. Only at this final compiler stage could we verify if the hardware accepted the architecture entirely in silicon, without leaving any layers as pure software fallbacks, and whether the QAT accuracy would survive the physical mapping without being destroyed by silent quantization errors.

IV. RESULTS AND DISCUSSION

In this section, we present and analyze the experimental results obtained across the different stages of the development pipeline: the baseline Keras floating-point execution, the naive quantization step, the Quantization-Aware Training (QAT) fine-tuning, and finally, the hardware-in-the-loop execution on the physical Brainchip Akida 1000 neuromorphic board.

A. Quantitative Performance Analysis

To evaluate the impact of the strict hardware constraints and the low-bitwidth quantization on the system's performance, the evaluation metrics were tracked at each milestone. Table I summarizes the parameters, losses, accuracies, and real hardware throughput (expressed in frames per second) for both the 7-class (Small) and the 18-class (Big) scenarios.

The empirical data highlights several critical phenomena regarding hardware-constrained quantization:

- **The Quantization Shock:** A direct, naive conversion of the Keras float32 weights into low-bitwidth integer representations without any prior optimization triggered a catastrophic collapse in performance. The accuracy dropped dramatically to 20.58% for the Small model and

14.95% for the Big model. This severe degradation validates our analysis in Section III: the 4-bit constraint on the internal convolutional layers and the clipping effects resulting from standard weight distribution conversions heavily distort the network's representative capacity if left unmanaged.

- **Efficacy of QAT and Bitwidth Scaling:** By applying Quantization-Aware Training, the network learned to compensate for the reduced bit-width resolution. It is worth noting that following Brainchip Akida v1 specifications, the first convolutional layer utilized an 8/4/4 configuration to preserve the sensory signal integrity coming from the high-frequency accelerometer and gyroscope data. The subsequent hidden layers operated strictly under a 4/4/4 bitwidth topology. Through patient QAT fine-tuning, the accuracy was successfully recovered to 91.18% (Small) and 72.81% (Big).
- **Hardware Validation (Simulation vs. Physical NPU):** Upon compiling the models into the native Akida Spiking Neural Network format (`.fbb`) and uploading them to the physical AKD1000 SoC, the hardware execution yielded 91.41% for the Small scenario and 72.82% for the Big scenario. It can be tentatively concluded that the software simulator provided by the Akida execution framework perfectly models the structural and physical constraints of the silicon, resulting in zero accuracy degradation during the final on-chip deployment.

B. Literature Benchmarking and Discussion

The performance of the 7-class Small scenario provides a direct anchoring point against previous literature. In the reference study conducted by Fra et al. [3], a similar SNN architecture operating on the same WISDM subset reported an accuracy of approximately 92.5%. Our physical hardware implementation achieved 91.41%, exhibiting a minimal drop of roughly 1.09%. This narrow margin is an exceptionally positive result, considering that our network had to be aggressively restricted to fit hardware constraints.

Comparing Memory Footprint we achieved a very low Memory occupation using only 0.54 MB of space for the Small (7-class) scenario against the 0.67MB achieved by Fra et al. [3].

Furthermore, the physical execution statistics reported remarkable computational throughputs. The Small model processed 2,012.03 frames per second, while the Big model reached 2,649.21 frames per second. This high framerate is inherently tied to the event-driven, sparse activation paradigm of neuromorphic processing, where computation is triggered asynchronously only upon spike generation. This high effi-

ciency suggests that the proposed pipeline is highly sustainable for continuous, ultra-low-latency on-edge classification in wearable health and AIoT monitoring applications.

C. Behavioral and Error Analysis

To investigate the classification behavior and identify typical misclassifications across the different physical movements, the confusion matrices for both configurations were generated from the hardware inference phase.

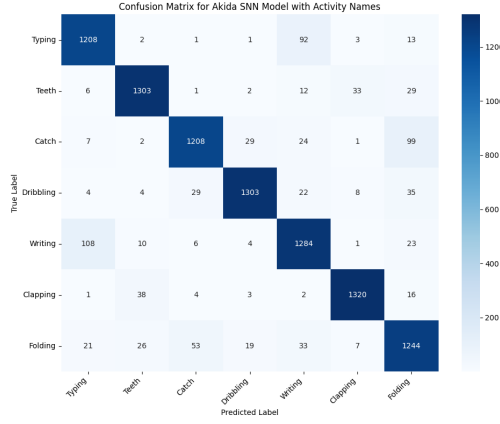


Fig. 5. Confusion Matrix for the 7-class (Small) scenario executed on the physical Akida 1000 hardware.

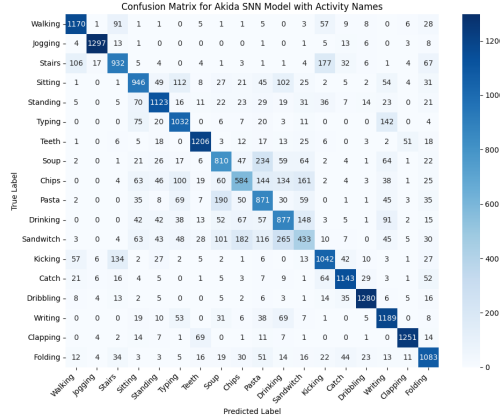


Fig. 6. Confusion Matrix for the 18-class (Big) scenario executed on the physical Akida 1000 hardware.

As illustrated in Figure 5, the 7-class configuration demonstrates strong class separation, with most hand-oriented activities showing high true positive rates. The small accuracy degradation compared to the baseline is mostly scattered across closely related physical movements.

Conversely, the 18-class scenario shown in Figure 6 reveals a more challenging environment. The drop in accuracy to 72.82% indicates that when the label space expands to incorporate more fine-grained or overlapping activities, the expressive capacity of a highly-constrained 4-bit network begins to saturate. It appears that specific groups of activities share similar sensory frequency signatures on the accelerometer and gyroscope axes, making them harder to disentangle under severe

limits and quantized weight boundaries. Nevertheless, given the strict zero-external-memory footprint, the model maintains a highly competitive trade-off between edge autonomy and analytical accuracy.

V. CONCLUSIONS AND FUTURE WORKS

In this report, we successfully demonstrated the viability of deploying complex Machine Learning algorithms for Human Activity Recognition (HAR) on ultra-low-power neuromorphic hardware. By addressing the severe limitations of edge computing, we developed a complete Hardware-Software Co-design pipeline to map a Convolutional Neural Network onto the physical Brainchip Akida 1000 SoC.

Our methodology highlighted the critical challenges of adapting standard Deep Learning architectures to physical silicon constraints. By meticulously managing the input padding logic, completely removing Batch Normalization to prevent 4-bit weight clipping, and leveraging Quantization-Aware Training (QAT), we achieved a seamless transition from a floating-point CNN Keras simulation to a fully quantized event-driven Spiking Neural Network (SNN).

Tested directly on the physical Akida hardware using the WISDM dataset, the system achieved a 91.41% accuracy on the 7-class benchmark scenario at a highly efficient throughput of 2013 FPS and a Memory Footprint of just 0.54 MB. This result closely matches the 92.5% baseline established in prior neuromorphic literature [3], proving that massive edge energy optimization does not strictly imply a sacrifice in analytical accuracy.

A. Future Works

While the results on the 7-class subset are highly promising, the expansion to the 18-class scenario revealed an accuracy drop (to 72.82%), highlighting the saturation of the model's expressive capacity under the rigid architectural limits of the Akida 1.0 hardware. Therefore, several future research directions can be outlined:

- **Migration to Next-Generation Hardware:** The upcoming Brainchip Akida 2.0 architecture introduces native support for advanced layers. Porting this HAR pipeline to the 2.0 platform could allow the deployment of more complex, modern topologies without forced manual workarounds, likely restoring high accuracy for broader, multi-class scenarios.
- **Hardware-Level Power Profiling:** While the theoretical energy efficiency of event-based inference has been confirmed by the high framerates, future works should involve direct physical power profiling. Using hardware probes to measure the exact milliwatt (mW) consumption during physical inference would allow for a strict, quantitative benchmarking of the energy savings against traditional Edge CPUs and conventional Microcontrollers executing standard ANNs.
- **Alternative Data Segmentation and Subject-Independent Evaluation:** While the current methodology relies on fixed 2-second temporal windows, future research will explore different window durations to

analyze the trade-off between inference latency, memory footprint, and the network’s ability to capture complex macroscopic activities. Furthermore, to rigorously validate the model’s generalization capabilities, we propose adopting a subject-independent data split strategy. By partitioning the training, validation, and test sets strictly by user IDs, we can ensure the network is evaluated on entirely unseen individuals, thereby proving its robustness and preventing overfitting to the specific biomechanical signatures of the training subjects.

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